



Combined effect of stream drying and nutrient enrichment on macroinvertebrate community: experimental study from artificial stream mesocosms

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Received: 9 December 2021 / Accepted: 30 November 2022 / Published online: 10 December 2022
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Abstract

Ongoing climate change and rising water demands are resulting in the increasingly frequent occurrence of stream drying, particularly in smaller streams in humid temperate climates. These streams are often situated in outlying regions with insufficient wastewater treatment management leading to inconsistent inputs of nutrients into receiving water systems. Both stream drying and nutrient enrichment negatively affect local aquatic macroinvertebrate communities, but their combined effect is still unclear. This study aimed to investigate the combined effect of stream drying and nutrient enrichment on taxonomic and functional composition of aquatic macroinvertebrate communities. Towards this aim, an artificial stream system was used consisting of 36 mesocosms with aquatic macroinvertebrates. The mesocosms were exposed to four independent treatments based on different flow (perennial-drying) and nutrient (pristine-nutrient enriched) status. After 28 days of experiment, samples from different treatments were compared to evaluate combined effect of stream drying and nutrient enrichment. Despite expectations, no effect of drying on taxonomic and functional alpha diversity was observed, but beta diversity has significantly increased. Nutrient enrichment though caused to decrease in the alpha diversity indices, which resulted in high pre- and post-experiment samples dissimilarity within all treatments. Moreover, the combined effect of both stressors was antagonistic as stream drying moderated harsher impact of nutrient enrichment. Results emphasize an unexplored role of intermittent streams in nutrient cycling in temperate climates and highlight the need for further investigation of recently novel intermittent streams since they provide different environmental conditions for macroinvertebrates than perennial streams.

Keywords Mesocosm experiment · Stream intermittency · Nutrient enrichment · Aquatic invertebrates · Multiple stressors · Temperate climate

Introduction

Stream intermittency is a naturally prevalent phenomenon dominating in Mediterranean or arid regions (Datry et al. 2014; Leigh et al. 2016). However, it is increasingly frequent in more humid climates, such as Central Europe (Pařil

et al. 2019; Straka et al. 2019, 2021), due to the combined effects of climate change and rising water demands (Mantyka-Pringle et al. 2012; van Vliet et al. 2013; Ionita et al. 2017). Aquatic macroinvertebrates (MIV) inhabiting repeatedly drying stream reaches develop various mechanisms to persist in situ or avoid disturbance (Chester and Robson 2011; Storey and Quinn 2013; Bogan et al. 2017). Whereas the response of unadapted aquatic MIV to novel intermittent flow regime might differ from an expected abrupt loss of taxa (Bonada et al. 2007) and is within recent scientific research interests (Pařil et al. 2019; Crabot et al. 2021; Straka et al. 2021). Moreover, the impact of surface drying on aquatic MIV communities differ notably when combined with elevated nutrient levels originating from anthropogenic pollution (Stubbington et al. 2019).

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Nutrient enrichment—namely nitrogen and phosphorus—in streams occurs often in agricultural areas, accompanying a nonpoint source of organic pollution as a side-effect of extensive fertilizer usage (Uusitalo et al. 2000; Bennett et al. 2001; Carey and Migliaccio 2009). While in populated urban areas point source pollution input from sewage wastewater treatment plants (WWTP) dominate (Bennett et al. 2001; Mitsch et al. 2001; Passell et al. 2005). The optimal WWTP cleaning processes should consist of primary, secondary and tertiary treatments with secondary and tertiary treatments being essential for nutrient removal (Pagilla et al. 2006; Muga and Mihelcic 2008; Carey and Migliaccio 2009). However, according to European Environment Agency (EEA 2020), many WWTPs in Central Europe reach tertiary treatment predominantly around larger rivers surrounded by major cities with little focus on rural areas where smaller streams are prevailing. For such outlying regions, WWTP inputs cause excessive addition of nutrients into watercourses; e.g. the WWTP effluent may contain up to 50% surplus in nitrogen and phosphorus loads (Martí et al. 2009; Grantham et al. 2012) which may lead to eutrophication and connected degradation processes in the water recipients (Heathwaite 2010). In some previous studies, focused on elevated nutrients in intermittent streams (Rueda et al. 2002; Matthaei et al. 2010; Loskotová et al. 2021), a unimodal relationship between aquatic MIV taxa richness and elevated nutrient levels was observed. Aquatic MIV generally profit from higher nutrient input until a tolerance threshold is reached. From that point, taxa richness and abundance are continually decreasing. Nutrient enrichment can reduce instream faunal biodiversity by lowering taxonomic richness (Gafner and Robinson 2007; Everall et al. 2019). But the ability to thrive in nutrient-enriched streams is considered to be taxon-specific (Bonada et al. 2007; Dang et al. 2009; Crabot et al. 2021) since higher aquatic MIV densities or even higher taxa richness might be observed downstream of WWTP effluent, where the aquatic MIV assemblage is predominantly determined by its functional composition (Gücker et al. 2006).

It is still not evident how intermittent streams are affected by nutrient enrichment in comparison to perennial streams, i.e. if the drying period exacerbates or benefits ecosystem processing of increased nutrient inputs. Some previous studies (Andersen et al. 2004; Passell et al. 2005) show the degradation impact of wastewater effluents on the natural nutrient regime in intermittent streams. However, the impact on aquatic MIV communities during low flows or dry periods (Rueda et al. 2002; Grantham et al. 2012; Karaouzas et al. 2018) show inconsistent effects caused by multiple factors. Aquatic MIV in dry reaches are handicapped with complete surface water loss, but they are simultaneously not directly exposed to nutrient enrichment diluted in water. The influence of WWTPs in intermittent streams is thus an open

question since the regular wastewater effluents progressively change their hydrological status from intermittent to perennial (Rueda et al. 2002; Canobbio et al. 2009).

The aim of this research was to investigate the potential negative effects of stream drying and nutrient enrichment on the taxonomic and functional diversity of the aquatic MIV community. Since drying and nutrient enrichment were both established as severe stressors for aquatic MIV communities, it was hypothesized that their combined effect will amplify the negative effect on those communities, resulting in a harsher impact than each stressor acting separately. This hypothesis was tested under laboratory conditions in indoor artificial stream mesocosms. Aquatic MIV collected from a pristine perennial stream (i.e. community not adapted to either drying or nutrient enrichment) were used as a model. As WWTP effluents may contain many other potentially harmful substances (e.g. pharmaceuticals, detergents, heavy metals), only an artificially induced nutrient enrichment resembling untreated wastewater effluent concentrations was applied.

Materials and methods

Field sampling

Source community of aquatic MIV for this experiment originated from one perennial pristine stream Hidasi—3rd Strahler order stream located in Mecsek mountains, Hungary (46°11'43.9" N 18°19'06.1" E, 320 m a.s.l.), under temperate humid climate (Dfb) according to the Köppen–Geiger climate classification (Peel et al. 2007). Aquatic MIV were sampled in April 2019 using the quantitative method; 1 min active disturbing a streambed within Hess sampler (250 µm mesh-size net, 16 cm in diameter, sampling area 200 cm²). Altogether 72 MIV samples were collected and pairs of samples (closest spatially) were then pooled. The resulting 36 pooled samples were then each split into two equal subsamples. First half was immediately preserved with 4% formaldehyde, where invertebrates were sorted under stereomicroscope, with these data being used to characterize aquatic MIV composition before the experiment (pre-experiment phase; PRE subsample). Second half-subsample was transferred in a plastic box with stream water to an indoor artificial stream system and inserted into prepared mesocosms (POST subsample). Five additional samples were split into halves in situ and their aquatic MIV composition compared to detect a possible splitting error.

Mesocosms in indoor artificial stream system

The indoor artificial stream system, located at the Department of Hydrobiology, University of Pécs, had four

independent channels (0.8 m wide and 2.4 m long) where dechlorinated tap water was circulated (velocity 5 cm s^{-1}) using three automatic pumps (AL-KO Drain 11001) per channel to imitate a stream flow (Fig. 1; for more details see Pernecker et al. 2020, 2021). Each channel was filled with tap water to a capacity of 0.9 m^3 . Nine experimental polyethylene cone mesocosms (height 20 cm, diameter 15 cm; Fig. 2a), with the same area as Hess sampler, were deployed within each channel. In total, 36 mesocosms with aquatic MIV samples were obtained. Each mesocosm had side perforations on the polyethylene base and four circular holes

(7 cm in diameter) along the whole perimeter filled with 0.5 mm mesh netting to enable current flow but prevent the escape of aquatic MIV. The top was covered with a dome-shaped strainer (1 mm mesh netting) to avoid emerging terrestrial adults from escaping (see Figs. 1 and 2a). Each mesocosm was semi-filled with a sterile inorganic substrate (mixture of gravel of 2–4 mm and 6–8 mm grain size).

Four channels represented four different treatments based on flow and nutrient status: (1) perennial-pristine (PER-PRIST), (2) drying-pristine (DRY-PRIST), (3) perennial-nutrient enriched (PER-NUTEN) and (4) drying-nutrient enriched (DRY-NUTEN).

To stabilise water chemistry, water was permanently circulated for seven days in all channels with installed mesocosms, then the aquatic MIV subsamples were inserted. Constant air temperature of 18°C was set and actual outdoor photoperiod was maintained using a room air conditioner and daylight white (4000 K) LED tubes with an automated timer. Dried leaves from the sample site were added to all mesocosms as a food source for the aquatic MIV and as a platform for microbial processes. Nutrient enrichment—represented by solid chemical compounds with the following concentrations added $\text{C}_6\text{H}_{12}\text{O}_6$ (350 mg L^{-1}), NH_4Cl (191 mg L^{-1}), KH_2PO_4 (87.7 mg L^{-1}) and KNO_3 (289 mg L^{-1})—was diluted in two channels (PER-NUTEN and DRY-NUTEN) a day before the aquatic MIV were added. Water level in two channels (DRY-PRIST and DRY-NUTEN) was suddenly lowered two days later under the base of the mesocosms (see Fig. 2b) where water has been kept circulating. An experimental period of 28 days was followed by sorting of aquatic MIV from all mesocosms, fixed (4% formaldehyde) and identified to the lowest possible taxonomic level (predominantly genus or species) representing a post-experiment phase (POST). Dissolved oxygen concentration, oxygen saturation, conductivity and pH, were measured weekly in each channel along with collecting emerging adults from each mesocosm. Water chemical analyses for biological oxygen demand (BOD_5), N-NH_4^+ , N-NO_3^- and P-PO_4^{3+} were performed in each treatment three times during the experiment. Means of all measured values for the variables are summarized in Online Resource 1.

Data analyses

All analyses were conducted using R 3.6.3. statistical software (R Core Team 2020), with outcomes presented as interaction plots using *ggplot2* package (Wickham 2016). Possible splitting error of five additional aquatic MIV samples was tested using one-way ANOVA to allow evaluation of matching PRE and POST subsamples in further analyses. The effect of drying, nutrient enrichment and their interaction on measured physical–chemical variables was tested using two-way ANOVA to assess the primary assumption



Fig. 1 Detail of one of four artificial channels equipped with automatic pumps and rubber hoses to imitate a stream flow. Nine experimental mesocosms for an independent treatment are installed within the channel

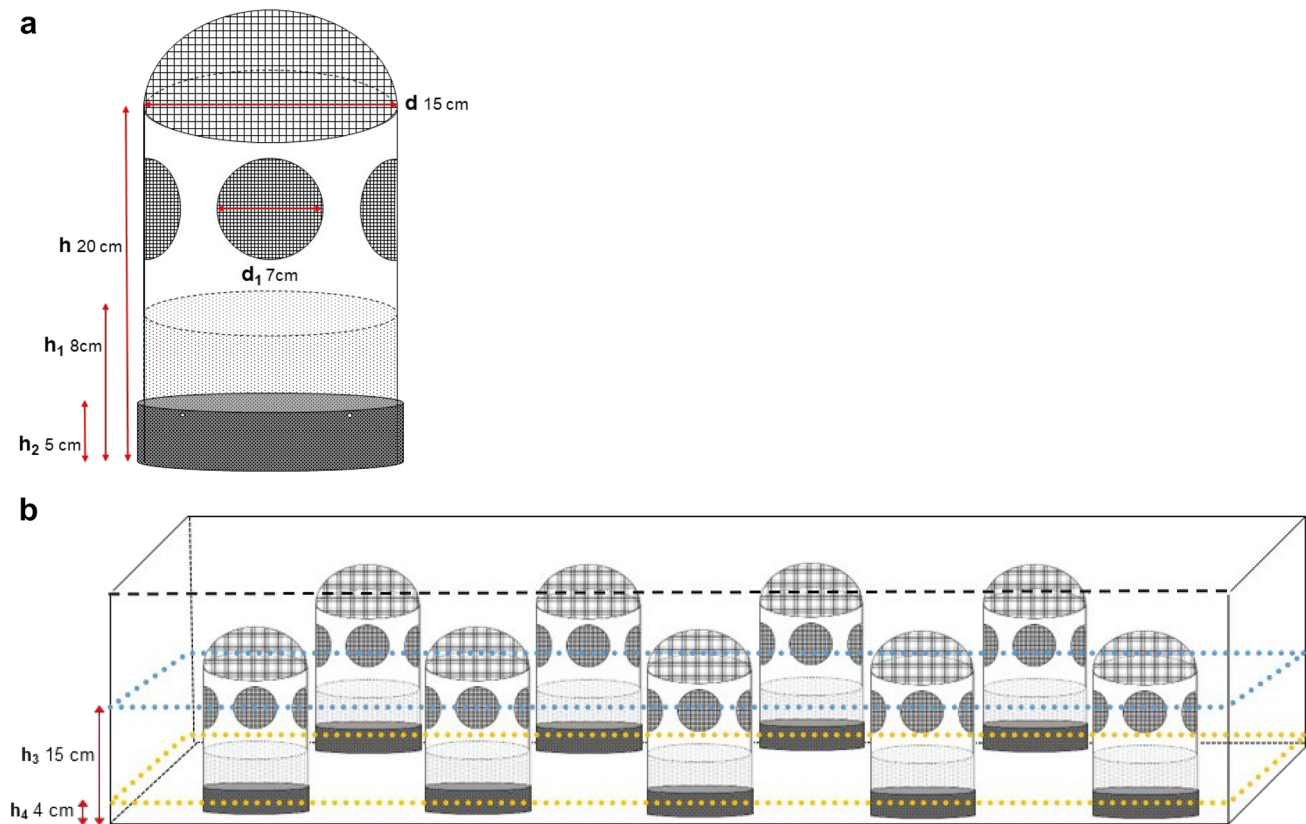


Fig. 2 **a** Scheme of polyethylene cone mesocosm covered with dome-shaped strainer; h = height of mesocosm, h_1 = height of inorganic substrate, h_2 = height of the base, d = cone diameter, d_1 = circular hole diameter; **b** scheme of an experimental channel; h_3 = blue dotted line

represents water level in perennial treatments (PER), h_4 = yellow dotted line represents water level in drying treatments (DRY) after flow cessation

that the addition of a chemical compound mixture affects the water quality. Next, the response of alpha-taxonomic indices (species richness and Shannon Diversity Index) to both stressors and their interaction were tested using their differences between PRE and POST subsamples to express the alpha diversity loss in individual treatments. The loss of species and decrease in Shannon diversity index were fitted using the most parsimonious generalized estimating equations (GEE) considering the position of mesocosms in the treatments to remove possible autocorrelation (using *geepack* package; Højsgaard et al. 2006). The beta-diversity (i.e. dissimilarity between matching PRE and POST subsamples) was calculated as Jaccard Distance and Bray–Curtis Dissimilarity Index using *vegan* package (Oksanen et al. 2019). The effect of drying, nutrient enrichment and their interaction on both beta-diversity indices was tested using two-way ANOVA.

For trait-based diversity calculation, the following nine biological traits were selected from a publicly available national database (<http://hydro.chmi.cz/riverchange/>) and Tachet trait database in Schmidt-Kloiber and Hering (2015): potential number of reproduction cycles per year,

reproduction type, dispersal strategy, respiration, locomotion type, substrate preference, feeding habit, armouring and body flexibility. Traits were selected based on their ecological relevance (see Datry et al. 2017) and sufficient data completeness for taxa present in the experiment as a criterion. For functional composition, larvae and adults of aquatic beetles were considered different taxa. The functional richness (based on Villéger et al. 2008) was calculated using *FD* package (Laliberté et al. 2014) and its response to drying, nutrient enrichment and their interaction, expressed as functional richness loss, was likewise tested using the GEE model.

Results

Together 13,056 individuals representing 91 different taxa were recorded in the whole study: 11,911 individuals from 85 taxa before (PRE) and 1145 individuals from 50 taxa after the experiment (POST). Taxa densities and frequencies are listed in Online Resource 2. No significant splitting error in five additional samples was detected

meaning both matching subsample halves were in fact equal in terms of taxa presence ($p = 0.862$) and abundance ($p = 0.689$), legitimising mutual comparison of sample halves (PRE and POST subsample) in further analyses. Added chemical compounds significantly increased levels of conductivity and the nutrient compounds (N-NH_4^+ , N-NO_3^- and P-PO_4^{3+}), while dissolved oxygen concentration and its saturation were significantly decreased in both nutrient-enriched treatments (PER-NUTEN, DRY-NUTEN). Also no significant difference was found in BOD_5 and pH among treatments (see Table 1).

A significant decrease in species richness (Fig. 3a) and Shannon Diversity Index (Fig. 3b) (both $p < 0.001$) in nutrient-enriched treatments, with no single effect of drying ($p = 0.446$ for species richness and $p = 0.331$ for Shannon Diversity Index) was observed. However, the interaction between drying and nutrient enrichment was marginally significant ($p = 0.071$) for species richness and highly significant ($p = 0.002$) for the Shannon diversity index. Drying moderated effect of nutrient-enriched treatment when combined.

Regarding beta-diversity, the significant dissimilarity between matching PRE and POST subsamples caused by nutrient enrichment expressed as Jaccard Distance ($p < 0.001$; Fig. 3c) and Bray–Curtis Dissimilarity Index ($p = 0.009$; Fig. 3d) was detected. The effect of drying ($p = 0.017$ and $p = 0.005$ for Jaccard Distance and Bray–Curtis Dissimilarity, respectively) and interaction of both stressors ($p = 0.011$ and $p = 0.009$ for Jaccard Distance and Bray–Curtis Dissimilarity, respectively) was also significant (see Fig. 3c, d). Similar to alpha diversity indices, drying mitigated the increase of beta diversity when both stressors occurred together.

Functional richness significantly decreased with nutrient enrichment ($p < 0.001$), with no significant effect of drying ($p = 0.669$) as well as the interaction of those two stressors ($p = 0.623$; Fig. 3e).

Discussion

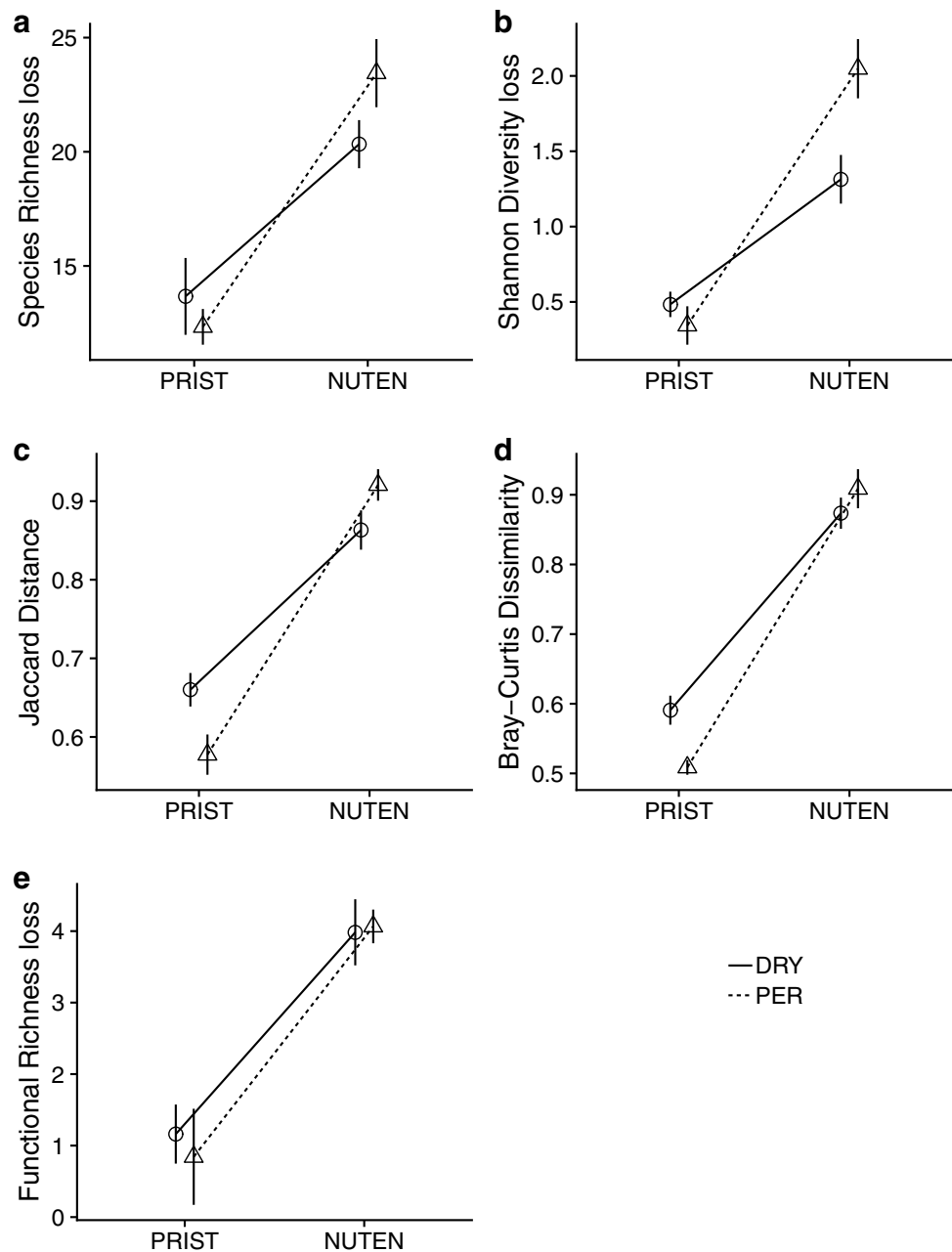
Contrary to expectation, a combination of stream drying and nutrient enrichment had a less pronounced negative effect than the influence of simple nutrient enrichment on the unadapted aquatic MIV community. Other experimental studies (Matthaei et al. 2010; Elbrecht et al. 2016) demonstrated either synergistic or no interaction of drying and elevated nutrient enrichment on aquatic MIV. However, in this study, an interesting result arose—an antagonistic interaction between those two stressors. While acting simultaneously, impact of nutrient enrichment was moderated by drying, so their combined effect was less harsh to the aquatic MIV than the impact of nutrient enrichment itself. As sub-surface flow was simulated, water could rise by capillary action which kept the stream sediment moist, enabling some aquatic MIV to survive three weeks of drying as previously observed (Stubbington and Wood 2013; Vadher et al. 2017; Loskotová et al. 2019). Moisture retention provides milder conditions during stream drying enhancing aquatic MIV survival (Verdonschot et al. 2015; Pařil et al. 2019; Loskotová et al. 2021). In addition, the absence of flowing surface water reduces direct contact of aquatic MIV with elevated nutrients (e.g. through tegument or gills) and thus limits their immediate negative impact.

When stream ecosystems are influenced by other natural or anthropogenic factors (e.g. chemicals, morphological degradation), aquatic MIV communities could be already stressed with those factors, so a response to a newly appearing stressor might not be as pronounced (Gücker et al. 2006; Heathwaite 2010; Grantham et al. 2012). For example, the well-known response of aquatic MIV to stream drying represents an ability to enter the inactive dormant stage (sensu Strachan et al. 2015) protecting from complete drying out. This experiment revealed a similar trend—if aquatic MIV are able to withstand a single stressor, the effects of other stressors are barely pronounced even if these communities were not previously selected by the environmental conditions. Acting separately, the effect of nutrient enrichment

Table 1 Estimated mean effects and p values in two-way ANOVA for both stressors (DRY = drying and NUTEN = nutrient enrichment) and their interaction on measured physical–chemical parameters; n.s. = not significant

	DRY		NUTEN		DRY $\times$ NUTEN	
	Estimate	p	Estimate	p	Estimate	p
Conductivity ($\mu\text{S cm}^{-1}$)		n.s.	1188.8	< 0.001		n.s.
N-NH_4^+ (mg L^{-1})		n.s.	38.55	0.008		n.s.
N-NO_3^- (mg L^{-1})		n.s.	358.21	0.004		n.s.
P-PO_4^{3+} (mg L^{-1})		n.s.	85.77	< 0.001		n.s.
Oxygen (mg L^{-1})		n.s.	− 2.77	< 0.001	− 2.25	< 0.001
Oxygen (%)		n.s.	− 26.46	< 0.001	− 23.71	< 0.001
BOD_5 (mg L^{-1})		n.s.		n.s.		n.s.
pH		n.s.		n.s.		n.s.

Fig. 3 The interaction plots comparing corresponding PRE and POST experiment subsamples (mean $\pm$ standard error) in four experimental treatments: **a** species richness and **b** Shannon Diversity Index significantly decreased (expressed as increased losses) in nutrient-enriched treatments, in interaction with drying, but with no single effect of drying. Both **c** Jaccard Distance and **d** Bray–Curtis Dissimilarity show significant differences for single effects of nutrient enrichment, drying and their interaction. With nutrient enrichment significantly decreased **e** functional richness (increased its losses, respectively), with no single effect of drying or interaction of both stressors. Nutrient status is marked on X axis (PRIST = pristine, NUTEN = nutrient enriched), flow status is represented by different symbols (circle = drying, triangle = perennial) connected with different line types (solid = DRY = drying, dashed = PER = perennial)



was intensified, whereas the effect of drying was rather discreet, indicating elevated nutrient enrichment as a more dominant stressor controlling the aquatic MIV community structure over stream intermittency. In some field studies (Kalogianni et al. 2017; Karaouzas et al. 2018; Arenas-Sánchez et al. 2021) enhanced negative effect of intermittency and nutrient inputs from agricultural and urban areas on aquatic MIV was observed, however, with a more pronounced aquatic MIV response to nutrient enrichment and milder effects of changes in hydrological regime. In addition, these studies were conducted in a Mediterranean region with assumed deviations in aquatic MIV response compared to temperate regions where no such study has

been conducted thus far. Moreover, aquatic MIV samples were collected mainly during wet phases here (Kalogianni et al. 2017; Karaouzas et al. 2018), with different environmental conditions than after drying following flow cessation. Changes in functional richness were in this experiment completely determined by nutrient enrichment, where sensitive taxa were almost eliminated regardless of flow regime suggesting the overwhelming influence of increased nutrients on specific trait composition in aquatic MIV community. This result corresponds with the above-stated effect of co-influence of a dominant and marginal stressor and is in line with an observed negative effect of nutrient enrichment on sensitive taxa (Arenas-Sánchez et al. 2021).

These findings could be utilized in rural areas with WWTPs without secondary or tertiary treatment, where generally smaller intermittency-prone streams are located. There, water management should focus its efforts primarily on effective nutrient removal from wastewater and prevent episodically pulsed point source pollution (e.g. overloads of WWTP with stormwater), according to EU Water Framework Directive 2000/60/EC (European Commission 2000). These actions can lead to the improvement of the ecological status of streams even if the water scarcity stressor prevails. In fact, nutrients tend to deposit in sediments during stream drying and return to the nutrient cycle with the next rewetting (von Schiller et al. 2017; Arce et al. 2019) causing an excessive flux in nutrient loads. To improve current knowledge of biochemical processes in intermittent streams, this follow-up process of rewetting should be considered to further evaluate the potential role of the dry streambed and nutrient regime effects on aquatic MIV community (Karaouzas et al. 2018; Stubbington et al. 2019). In that research, the climatic context has to be regarded as well, since different climate zones (e.g. Mediterranean, temperate) establish different long-term environmental conditions. Thus, from an evolutionary perspective, biotic communities could respond inconsistently to similar interactions of stressors, indicating that patterns revealed in one climate zone could not be easily transferable to another (Bonada et al. 2007; Crabot et al. 2021).

Conclusion

If a stream is suddenly affected by point source pollution, accompanied by nutrient enrichment (e.g. WWTP effluent), its aquatic MIV community, not adapted to these conditions, rapidly decreases in terms of taxa richness, abundances and even functional richness. Nutrient enrichment and water pollution can have a much stronger negative effect on MIV communities than stream drying, which could surprisingly immediately mitigate their harsh impact. In a long-term perspective, however, stream intermittency with its consequences yet represents negative disturbance as well. These statements might be primarily considered in water management of outlying regions with limited high-efficiency WWTP, where ongoing climate change and increasing water demands trigger more frequent stream intermittency along with not yet implemented adequate solution of agricultural and urban nutrient input removal from watercourses.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00027-022-00924-w>.

Acknowledgements This study was supported by the H2020 European Research and Innovation action Grant Agreement no. 869226 (DRYvER) and the Czech Science Foundation (P505-20-17305S). BP

and ZC were supported by the Higher Education Institutional Excellence Programme of the Ministry of Human Capacities in Hungary (20765-3/2018/FEKUTSTRAT, TUDFO/47138/2019-ITM and 2020-4.1.1-TKP2020). We thank Balázs József Berta (University of Pécs, Hungary) and Denis Bučan (University of Zagreb, Croatia) for great help during fieldwork and set of experimental background and especially Éva Horváthné Tihanyi and Tímea Pernyeszi (both University of Pécs, Hungary) for providing chemical analyses and technical support. Special thanks goes to Jacob VanHouten (Delta College, US Michigan) for English proofreading.

Data availability The datasets generated during the research and analysed in the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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